Sahaj Bhadja

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Summary

Software Engineer with 2+ years building and operating **LLM-backed** services, **RAG** retrieval, model evaluations, and **ingestion pipelines** in **Python** across AWS, Azure, and GCP, from object storage to Kubernetes.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant async **LLM orchestration** with FastAPI background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs with 3x throughput and no cascading failures under concurrent load.
- Built **ingestion pipelines** across Blob Storage, PostgreSQL, and SQL Server with incremental loads and reconciliation, running 5K+ monthly jobs at 99%+ reliability, and tuned indexes cutting dashboard latency 71%.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built **LLM-backed APIs** across GPT-4o, Claude, and Qwen-VL behind a switchable provider layer with OCR fallback on timeouts, serving 150+ instructors and 10K+ daily grading requests on AWS.
- Tracked grading consistency and per-call token cost per model to guide default and fallback model routing, sustaining 90%+ **grading consistency** while cutting feedback turnaround 60% for instructors.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built end-to-end **data pipelines** processing 100K+ autonomous-driving trajectory logs and HD-map vectors into training features for an offline-RL GPT-based Decision Transformer, improving long-horizon planning 18% in simulation.
- Evaluated a spatial-temporal attention encoder against linear, Wayformer-, and MotionLM-style baselines in **closed-loop simulation**, using the results to adopt the HD-map-aware design that improved planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **ETL** ingestion on AWS Glue and S3 object storage, normalizing JSON, CSV, and Parquet data from multiple sources, boosting pipeline efficiency 68% and cutting storage costs 15%.
- Optimized large-scale API integrations with indexing, API-layer **caching**, pagination, and field filtering, improving response times 30%, and automated nightly Jenkins CI/CD with 20+ test suites across three platforms.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight Python **FastAPI** microservices with React and TypeScript front ends on Azure SQL, synchronizing 2M+ records across distributed locations through REST APIs and real-time WebSocket updates.
- Shipped Docker images to **Kubernetes** on Azure through Jenkins CI/CD with automated tests and approval gates, reaching 95%+ build validation and tracking production health with Sentry and Azure Monitor.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++
AI Systems: LLM APIs, RAG, Embeddings, FAISS, Evals, Prompt Chaining, Multi-Model Routing, PyTorch, Transformers
Backend & Data: FastAPI, Flask, REST APIs, Microservices, PostgreSQL, SQL Server, Object Storage, Caching, Async Job Queues
Pipelines: ETL/ELT, Incremental Loads, AWS Glue, Databricks, Spark
Cloud & Infrastructure: AWS, Microsoft Azure, GCP, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins
Frontend & Practices: React, Sentry, Monitoring, Cost Tracking, AI Coding Agents, Agile/Scrum

Projects

Agent and RAG Evaluation Harness | *Python, FastAPI, Claude, Embeddings, BM25, PostgreSQL, Docker*

- Designed a Python tool-calling agent over hybrid BM25 and embedding retrieval of technical documents, with a golden-task **eval** suite gating CI on retrieval recall, tool-call accuracy, and faithfulness.

Big-Brain Live | *LLaMA-2, FAISS, Embeddings, PostgreSQL, GCP, REST APIs*

- Built a **RAG** document assistant using LLaMA-2 with transformer embeddings and FAISS over 50K+ documents, with an incremental ingestion and indexing pipeline that cut retrieval latency 40% at 95% query accuracy.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, SQL Server, Docker*

- Built an AI-assisted SQL workspace in **Java** and TypeScript supporting 10+ databases, validating schema-aware AI-generated queries through structured execution to reach 82% accuracy against relational database engines.